

System Requirement Specification (SRS)

Project Title: Hostel Management & GatePass

Prepared By: Harshit Singh, Kshitiz Dixit, Aryan Gupta, Harsh Kasaundhan

Target Platforms:

- **Student:** Mobile App (iOS & Android)
- **Warden:** Mobile App (iOS & Android) + Web Dashboard
- **Security Guard:** Mobile App / Handheld Scanner
- **Super Admin:** Centralized Web Panel

1. Executive Summary & Business Objectives

This platform is a comprehensive, multi-tenant hostel management system designed to digitize student residency operations. By integrating a visual bed-booking engine, a dynamic QR-based gatepass workflow, and a centralized fee payment portal, the system eliminates manual administrative bottlenecks. The architecture connects multiple operational touchpoints: students, wardens, security personnel, and administrators into a unified, real-time database to ensure strict security, transparent financial tracking, and frictionless campus living.

Key Business Goals:

- **Automated Security:** Digitize campus entry/exit with highly secure, time-sensitive QR codes and immutable database logging.
- **Streamlined Accommodation:** Transition from manual room assignments to an interactive, user-driven bed selection model with strict administrative verification.
- **Centralized Finances:** Consolidate hostel, academic, and miscellaneous fee payments into a single, trackable in-app portal.
- **Real-Time Operations:** Utilize WebSocket architecture to deliver instant state changes for bookings, live security scans, and emergency push notifications.

2. User Roles, Onboarding & Access Control

2.1 Student (Mobile App)

- **Authentication:** University email/ID login with secure password and optional biometric app unlock.
- **Capabilities:**
 - Access the visual bed allocation matrix for individual or group bookings.
 - Initiate, track, and modify gatepass requests.
 - Communicate with wardens for emergency extensions.
 - View and pay outstanding hostel, academic, or miscellaneous fees.

2.2 Warden (Mobile App & Web Dashboard)

- **Authentication:** Secure staff credentials with role-based access.
- **Capabilities:**
 - Global view of all pending, active, and expired gatepasses.
 - Authority to approve, deny, or modify the timings of a student's gatepass request.
 - Receive automated push notifications for late arrivals and emergency extension requests.

- Direct in-app communication with students requesting schedule modifications.
- Finalize bed assignments upon physical verification of payment receipts.

2.3 Security Guard (Mobile App / Handheld Scanner)

- **Authentication:** Shift-based login with restricted access to scanning modules only.
- **Capabilities:**
 - Minimalist interface featuring a high-speed QR code scanner.
 - Instant on-screen validation displaying the student's name, photo, and pass validity.
 - Automated background logging of entry/exit timestamps, gate number, the specific guard on duty, and the warden who approved the pass.

2.4 Super Admin (Centralized Web Panel)

- **Authentication:** High-security login with multi-factor authentication.
- **Capabilities:**
 - Master Role-Based Access Control (RBAC) to provision, modify, or revoke permissions for any student, warden, or guard.
 - Full oversight of hostel inventory, floor plans, and billing cycles.
 - Access to comprehensive, exportable logs (security exits, fee transactions) for auditing and compliance.

2.5 User Onboarding & Identity Management

- **Hybrid Onboarding Pathway:**
 - **Admin Bulk Import:** Super Admins can upload CSV or Excel rosters at the beginning of the academic year to instantly provision accounts, mapping core fields and triggering welcome emails.
 - **Self-Registration:** Students can download the app and self-register. Accounts remain in "Pending Verification" with restricted permissions until approved.
- **Duplicate Detection & Conflict Resolution:**
 - The system uses the University Roll Number and registered Email as primary unique identifiers.
 - The Super Admin panel includes a "Data Integrity" module to review flagged duplicate profiles and execute a one-click **Merge** (consolidating logs/fees) or **Delete** (removing the erroneous duplicate).

3. Detailed Feature Specifications

3.1 Visual Bed Booking & Manual Assignment Workflow

- **Grid-Based Selection Interface:** A responsive CSS Grid / Flexbox UI displaying rooms and beds on a floor-by-floor basis. Available beds are clickable; occupied/locked beds are greyed out.
- **Soft Lock & Cart System:** Driven by WebSockets, selecting a bed applies a temporary, system-wide "soft lock" to prevent simultaneous bookings by other students. Supports multi-bed selection for group bookings.
- **Asynchronous Payment Flow:** Students lock the room and complete the payment through the in-app portal at their convenience.
- **Warden Physical Verification (Final Assignment):** The digital booking is logged as a

"Pending Assignment." The student must present the physical or downloaded PDF payment receipt to the Warden in person. The Warden then uses their dashboard to officially change the bed status from "Locked" to "Assigned."

3.2 Dynamic Gatepass & Security Verification Lifecycle

- **Request Initiation:** Students submit a gatepass request (reason, expected exit, and return time) via the mobile app.
- **Warden Review Queue:** Requests populate a shared dashboard accessible to all wardens. Wardens can modify the approved return time before issuing the final digital approval.
- **Dynamic QR Generation:** Upon approval, a secure QR code generates on the student's app. This QR code automatically expires exactly 60 minutes after the student's physical return is logged at the gate.
- **Guard Scanning & Immutable Logging:** Guards scan the code using a minimalist mobile app. A successful scan instantly queries the WebSocket server, displaying the student's credentials and logging the entry/exit timestamp, gate number, guard on duty, and approving warden ID into a secure PostgreSQL table.
- **Emergency Extensions & Late Alerts:**
 - Students delayed off-campus can use the app to message a warden and request a time extension.
 - If a student fails to return by the approved deadline, automated high-priority alerts are pushed in real-time to all wardens.

3.3 Unified Fee Management

- **Fee Dashboards:** Categorized visibility separating Hostel Fees, Academic Fees, and Miscellaneous Dues (e.g., fines, deposits).
- **Payment Gateway:** Integrated digital payment options (UPI, credit/debit, net banking) allowing asynchronous settlement.
- **Receipt Generation:** Successful payments instantly generate a downloadable PDF receipt, required for the physical bed assignment verification step.

4. Technical Architecture & Infrastructure

4.1 Core Tech Stack

- **Frontend (Web Portals):** React.js for the Super Admin dashboard and Warden interfaces, ensuring robust state management for live rosters and administrative controls.
- **Frontend (Mobile Apps):** React Native, maintaining a unified codebase deployed across distinct iOS and Android interfaces for Students, Wardens, and Security Guards.
- **Backend Services:** Node.js with Express, delivering concurrent RESTful APIs for standard CRUD operations (user management, billing).
- **Database:** PostgreSQL to maintain ACID compliance for financial transactions and strict relational mapping for gatepass and security logs.

4.2 Real-Time Infrastructure (WebSockets)

- **Persistent Connections:** WebSockets natively implemented (via Socket.io or AWS API Gateway) to handle all instantaneous state transitions.

- **Live Bed Locking:** Immediate broadcast to all active student clients when a bed is selected, updating the grid UI without page refreshes.
- **Instant Security Sync:** Sub-second latency for security QR scans, pushing validation to the guard's device and immediately updating the student's geographic state on the backend.
- **Live Push Alerts:** Real-time push notifications delivered directly to Warden dashboards for extension requests and automated late-return triggers.

4.3 Cloud Deployment & Storage

- **Infrastructure:** Containerized deployment using Docker on AWS to seamlessly scale backend instances during high-traffic periods (morning exits, semester starts).
- **Asset Management:** AWS S3 for secure, scalable hosting of static assets, including student profile images, graphical floor plans, and generated PDF fee receipts.

5. Summary Matrix of Decisions

Parameter	Selected Logic / Rule
Bed Booking UI	CSS Grid / Flexbox mapping; simple clickable room/bed cards (No canvas dependencies).
Booking Flow	Real-time WebSocket soft-lock cart system allowing asynchronous in-app payments.
Final Bed Assignment	Manual state change by Warden strictly upon physical verification of the payment receipt.
Gatepass Lifecycle	Mobile request → Shared Warden Review Queue → Approval/Modification.
Access Control	Encrypted QR Code expiring exactly 60 minutes after the return scan is logged.
Security Validation	Guard mobile app QR scanner with immutable PostgreSQL logging (Time, Gate, Guard, Approver).
Exception Handling	In-app extension requests and automated WebSocket-driven high-priority late alerts to Wardens.

Fee Management	Segregated ledgers (Hostel, Academic, Misc) with digital payment gateway integration and PDF receipt generation.
User Onboarding	Hybrid model: Super Admin bulk CSV upload + self-registration pathway.
Duplicate Handling	Super Admin "Data Integrity" module to manually flag, merge, or delete conflicting profiles.
Tech Stack	React.js, React Native, Node.js/Express, PostgreSQL.
Real-Time Data	WebSockets utilized from day one for instantaneous UI updates, locking, and alerts.